

CSE423 - Computer Graphics

Final Practice Sheet [Spring 2025]

N.B. This is merely a reference to the problems, inclusive and exclusive to the questions that will be set in the exam. This practice sheet does not include all kinds of questions that may come in the examination. The sole purpose of this practice sheet is to facilitate a clear and thorough understanding of the concepts.

1. Derive a 4x4 **simple purpose** perspective projection matrix using the appropriate figure, showing P, P', COP, PP. Also, this matrix can be converted into a simple perspective projection matrix. Possible cases are -
 - a. Origin is at COP and
 - b. Origin on the projection plane.
2. Derive a 4x4 **general-purpose** perspective projection matrix using the appropriate figure, showing PP', COP, PP.
3. For a point P (10, 20, -40), calculate the projected point P' using Orthographic Projection.
 - a. For the projection plane of XY.
 - b. For the projection plane of YZ.
 - c. For the projection plane of ZX.
 - d. For the projection plane of $y = -13$
4. For a point P (10, 20, -40), calculate the projected point P' using Cavalier Projection, orientation angle of 30° for the projection plane of XY.
5. For a point P (10, 20, -40), calculate the projected point P' using Cabinet Projection, where the orientation angle is 30°
 - a. For the projection plane of XY.
 - b. For the projection plane of $x = 7$.
 - c. For the projection plane of ZX.
6. For COP at origin, calculate the projected point P' for a given point P(50, 60, -300), if the plane is 200 units in the Z axis away from the COP.
7. For PP at origin, calculate the projected point P' for a given point P(30, 20, -100), if the plane is 200 units in the Z axis away from the COP.
8. Let a 3D point (423, -423, 423) be projected on a projection plane.

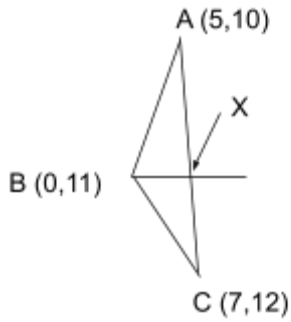
Given that the projection plane is at (0, 0, 400) and the coordinate of the COP is (4, 2, 3). Determine the coordinate of that 3D point on the projection plane using a general-purpose perspective projection matrix. How far is the Projection Plane (PP) from the Center of Projection (COP) in terms of units?

9. For COP at (100, 90, 0), calculate the projected point P' for a point P(30, 50, -250) where the Projection Plane (PP) is 200 units away from COP.
10. A 3D vertex P(40, 30, 70) is projected on the projection plane. Suppose near clipping plane distance, z_p (check whether it is 40). From the below general purpose perspective projection matrix;

$$\begin{bmatrix} 1 & 0 & -0.3333 & 13.3333 \\ 0 & 1 & 0.6667 & -26.6667 \\ 0 & 0 & -1.3333 & 93.3333 \\ 0 & 0 & -0.0333 & 2.3333 \end{bmatrix}$$

- a. Find out the Center of Projection (COP).
- b. How far is the Projection Plane (PP) from the Center of Projection (COP) in terms of units?
- c. Now, determine the projected coordinate P' on the projection plane.
- d. Can you identify the equations for this 4x4 projection matrix?
11. Let (50, 70, 100) be the coordinates of a light source of intensity 0.95 units. The light is illuminating a quad consisting of $P_0(10, 10, 5)$, $P_1(-10, 10, 5)$, $P_2(-10, -10, 6)$ and $P_3(10, -10, 6)$ vertices. Determine the intensity of the reflected light at the center of the quad using the diffuse reflection model. Given that the diffuse absorption coefficient of the quad surface is 0.8 units.
12. Let (30, 10, 500) be the coordinates of a light source of intensity 0.5 units. The light is illuminating a sphere whose center is at C(10, -15, 6). Determine the total intensity of the reflected light from a point P(20, 10, 120) on the sphere using the diffuse reflection model. Given that the diffuse absorption coefficient of the surface is 0.8 units.
13. A light source with an intensity of 15 and a radius of influence of 80 is located at (4,2,3) from which you are called to calculate the illumination of a point on the $y = 4$ plane. The camera is set at (2, 1, 5) and the light is reflected from points (3, 4, 4) of the plane. The ambient, diffuse, and specular coefficient is given at 0.4, 0.2, and 0.3. The shininess factor of the surface is 3. If the ambient light intensity is at 2, calculate the total reflected light intensity using Phong's Lighting Model.

14. Suppose there are two light sources in the scene. One light source is located at (4, 2, 3) with an intensity of 15 and a radius of influence of 20 and another one is located at (10, 40, 50) with an intensity of 5 and a radius of influence of 80. You are called to calculate the illumination of a point on the yz plane. The camera is set at (2, 1, 5) and the light is reflected from points (0, 3, 2) of the plane. The ambient, diffuse, and specular coefficients of the surface are given at 0.23, 0.7, and 0.5. The shininess factor of the surface is 5. If the ambient light intensity is at 6, calculate the total reflected light intensity using Phong's Lighting Model.
15. Write an algorithm for converting the RGB color values into HLS/HSB color values.
16. Write an algorithm for converting the RGB color values into HSV/HSB color values.
17. For a grayscale color with a green component value of 0.74, determine the corresponding values of the red and blue components.
18. For an achromatic color with a green component value of 0.74, determine the corresponding values of the red and blue components.
19. For a monochrome color with a green component value of 0.74, determine the corresponding values of the red and blue components.
20. Convert the RGB colors into both HLS/HSB and HSV color values.
 - a. (0.25, 0.3, 1.0)
 - b. (0.01, 1.0, 0.09)
 - c. (0.8, 0.8, 0.35)
 - d. (0,0, 0.4, 0.4)
 - e. (1.0, 1.0, 0.5)
 - f. (0.7, 0.71, 0.7)
 - g. (0.5, 0.5, 0.5)
 - h. (1.0, 1.0, 1.0)
 - i. (0.39, 0.398, 0.2)
21. Convert the HSV values into RGB color values: $H = 236^\circ$, $S = 0.3$, $V = 60\%$
22. Convert the HSV values into RGB color values: $H = 135^\circ$, $S = 0.9$, $V = 80\%$
23. Convert the HSV values into CMY color values: $H = 336^\circ$, $S = 0.7$, $V = 40\%$
24. To answer some of the following questions, you will need four variables A, B, C, and D, sequentially the first, second, third, and fourth pair of digits from the left in your student ID. For example, if your ID is 15101208, then $A = 15$, $B = 10$, $C = 12$, and $D = 8$.



- a. A color is given in CMY form with the values (0.A, 0.C, 0.D). Convert the color into an equivalent HSV model. Show the calculation in detail.
 - b. The color at vertex A is the value B of your student ID, and at vertex C is D of your student ID. Now, calculate the color at point X using Gouraud shading. Can a specular light on point X be captured using the above model? Why or why not?
25. Determine C(1) and G(1) continuity of the following functions at the given points:

- a. At $t = 2\pi$,

$$(x(t), y(t)) = \begin{cases} (t, \sin t) & \text{for } t \leq 2\pi \\ (t, 1 - \cos t) & \text{for } t > 2\pi \end{cases}$$

- b. At $t = \pi/4$,

$$(x(t), y(t)) = \begin{cases} (t, \sin t) & \text{if } t \leq \frac{\pi}{4} \\ (t, 1 - \cos t) & \text{if } t > \frac{\pi}{4} \end{cases}$$

- c. At $t = 1$,

$$(x(t), y(t)) = \begin{cases} (6t, t^3) & \text{if } t \leq 1 \\ (t^4 + 5, t^2) & \text{if } t > 1 \end{cases}$$

- d. At $t = 1$,

$$(x(t), y(t)) = \begin{cases} (t, t^2) & \text{for } t \leq 1 \\ (t, t^2 + (t - 1)^3) & \text{for } t > 1 \end{cases}$$

26. Find the explicit **representation** of a **quadratic** curve going through the following 3 points using the **Lagrange Polynomial**:
 $P_0 = (0, 0)$, $P_1 = (1, 2)$, $P_2 = (2, 0)$
27. Derive the **Basis Matrix** for the **cubic Bézier** curve.
28. Given four control points $P_0 = (1,1)$, $P_1 = (2,3)$, $P_2 = (4,3)$, and $P_3 = (5,1)$, find the point on the **cubic Bézier** curve at $t = 0.5$.
29. Given the four control points in 3D:
 $P_0 = (0,0,0)$, $P_1 = (3,6,0)$, $P_2 = (6,6,6)$, $P_3 = (9,0,6)$
 Find the point on the **cubic Bézier** curve at $t = 0.5$.
30. Given the four control points in 2D:
 $P_0 = (0,0)$, $P_1 = (2,2)$, $P_2 = (4,-2)$, $P_3 = (6,0)$
 Find the point on the **cubic Bézier** curve at $t = 0.75$.
31. Given the first three control points of a cubic **Bézier** curve:
 $P_0 = (2, 1)$, $P_1 = (3, 4)$, $P_2 = (5, 6)$
 and the point on the curve at $t = 0.5$:
 $f(0.5) = (4, 5)$
 Find the **fourth control point**, $P_3 = (x_3, y_3)$.
32. You are going to draw 3 cubic **Bézier** curves joined together to form a single smooth composite curve. You have already decided upon the control points for the first and last Bézier curves:
- First Bézier curve (Curve A):
 $A_0 = (0, 0)$, $A_1 = (1, 2)$, $A_2 = (2, 2)$, $A_3 = (3, 0)$
 - Third Bézier curve (Curve C):
 $C_0 = (6, 0)$, $C_1 = (7, -2)$, $C_2 = (8, -2)$, $C_3 = (9, 0)$

You want to insert a **Bézier curve (Curve B)** between them such that the entire 3-curve segment is **C(1) continuous**.

Find the **4 control points**- B_0 , B_1 , B_2 , B_3 of the **middle Bézier** curve (Curve B).